IGCSE 0607 Practice Questions — E2.1 — Combined Questions & Markscheme

Introduction to algebra

Total Questions: 21 — **Total Marks:** 94 Marks

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- **E2.1** Introduction to algebra
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

1 Find the value of $3a - 2b$ when $a = 4$ and $b = -5$.

..... **[2]**

2 Given the formula $v = u + at$.

Find the value of v when $u = 5$, $a = -2$, and $t = 3$.

$v =$ ..... **[3]**

3 **(a)** Work out the value of $4x - 3y$ when $x = -2$ and $y = 5$.

..... **[2]**

(b) Work out the value of $\dfrac{p^2 - 7}{2q}$ when $p = 5$ and $q = -3$.

..... **[2]**

4 **(a)** Work out the value of $5x - 2y$ when $x = 3$ and $y = -4$.

..... **[2]**

(b) Evaluate

$$\frac{a^2 - b^3}{2c}$$

when $a = -4$, $b = 2$ and $c = 3$.

..... **[2]**

5 (a) **(a)** The cost of hiring a plumber is calculated using the formula

$C = 40 + 25h$ where C is the cost in dollars and h is the number of hours worked.
Work out the cost of hiring a plumber for 3.5 hours.

$C = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

(b) Given that $v = u + at$, find the value of v when $u = -12$, $a = 9.8$ and $t = 3$.

$v = \ldots\ldots\ldots\ldots\ldots$ **[3]**

6 The distance, d, between two points (x_1, y_1) and (x_2, y_2) is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

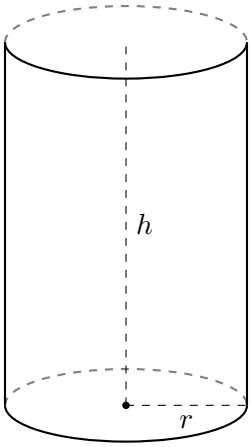
(a) Work out the value of d when $x_1 = 4$, $y_1 = -3$, $x_2 = -8$, and $y_2 = 2$.

$d =$ ..... **[2]**

(b) Given that $d = 17$, $x_1 = k$, $y_1 = 8$, $x_2 = 3$, and $y_2 = -7$, work out the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

7 A cylindrical container of radius r and height h is open at the top.
The total exterior surface area of the container, A, consists of the curved surface area and the circular base.



(a) Show that the total surface area A can be written as

$$A = \pi r(2h + r)$$

[2]

(b) **(i)** Calculate the exact surface area A when $r = 4$ and $h = 6$.
Give your answer in terms of π.

$A =$ ..... **[2]**

(ii) The volume of the cylinder is V.
The formula for the volume is $V = \pi r^2 h$.
Express h in terms of V, π and r, and substitute this into the expression for A to show that

$$A = \frac{2V}{r} + \pi r^2$$

[3]

8 Consider the expression $P = \dfrac{2x^2 - 3y}{x - y}$.

(a) Calculate the value of P when $x = -2$ and $y = 4$.

$P =$ ..... **[3]**

(b) Rearrange the formula to make y the subject.

$y =$ ..... **[4]**

9 The volume, V, of a cone with radius r and height h is given by the formula

$$V = \frac{1}{3}\pi r^2 h$$

(a) Calculate the value of V when $r = 6.4$ and $h = 12.6$.
Give your answer correct to 1 decimal place.

$V =$ ..... **[2]**

(b) Work out the value of r when $V = 800$ and $h = 14$.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

10 The cost, C dollars, of hiring a plumber for h hours is given by the formula:

$C = 45 + 35h$

(a) Calculate the cost of hiring the plumber for 3.5 hours.

$C =$ ..... **[2]**

(b) The plumber charges a total of \$325 for a job.
Calculate the number of hours the plumber worked.

$h =$ ..... hours **[3]**

11 The volume, V, of a spherical cap of height h and base radius a is given by the formula

$$V = \frac{1}{6}\pi h(3a^2 + h^2)$$

(a) Calculate the volume of a spherical cap with height $h = 4.2$ cm and base radius $a = 6.5$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm^3 **[2]**

(b) Rearrange the formula to make a^2 the subject.

$a^2 =$ ..... **[3]**

(c) Find the base radius, a, of a spherical cap with volume $V = 150$ cm^3 and height $h = 3$ cm. Give your answer correct to 3 significant figures.

$a =$ ..... cm **[1]**

12 A formula used in physics is

$$s = ut + \frac{1}{2}at^2$$

(a) **(a)** Calculate the value of s when $u = 5.2$, $t = 4.5$ and $a = -9.8$.

$s =$ ..... **[2]**

(b) Calculate the positive value of t when $s = -9.8$, $u = 4.9$ and $a = -9.8$.

$t =$ ..... **[3]**

13 Work out the value of

$$\frac{3x - y^2}{2z}$$

when $x = 5$, $y = -4$ and $z = 0.5$.

..... **[2]**

14 The cost, C dollars, of hiring a campervan for d days and travelling m kilometres is given by the formula

$$C = 85d + 0.24m$$

(a) Calculate the cost of hiring the campervan for 7 days and travelling 450 kilometres.

$ ..... **[2]**

(b) A family hires the campervan for 14 days.
The total cost is $1475.60.
Work out the number of kilometres they travelled.

..... km **[3]**

15 The mathematical operation $*$ is defined for all non-zero numbers a and b as

$$a * b = \frac{a^2 - b^2}{ab}$$

(a) Evaluate the following, giving each answer as a fraction in its simplest form.

(i) $6 * (-2)$

..... **[2]**

(ii) $\frac{1}{2} * 4$

..... **[3]**

(b) Given that $x * 2 = \frac{3}{2}$, find the positive value of x.

$x =$ ..... **[3]**

16 Work out the value of

$$5x^2 - 3y$$

when $x = -2$ and $y = 4$.

..... **[2]**

17 **(a)** A plumber charges a fixed fee of \$45 plus \$32 for each hour worked.
Write down a formula for the total charge, C dollars, when the plumber works for h hours.

$C =$ ..... **[1]**

(b) Work out the total charge when the plumber works for 3.5 hours.

\$ ..... **[1]**

(c) Work out the value of

$$x^3 - 4xy$$

when $x = -3$ and $y = 2.5$.

..... **[2]**

18 The value of y is given by the formula

$$y = \frac{a - b^2}{c - \sqrt{d}}$$

Work out the value of y when $a = -4.5$, $b = -3.2$, $c = 1.8$ and $d = 0.64$.

$y =$ ..... **[3]**

19 **(a)** Work out the value of $3a - 5b$ when $a = -2$ and $b = 4$.

..... **[2]**

(b) Work out the value of $x^2 + 4x$ when $x = -3$.

..... **[2]**

20 A solid shape has a volume, V, given by the formula

$$V = \pi r^2 h + \frac{2}{3}\pi r^3$$

Work out the value of V when $r = 4.5$ and $h = 8.2$.
Give your answer correct to 3 significant figures.

$V =$ ..... **[3]**

21 **(a)** A plumber charges a fixed fee of \$40 plus \$25 for each hour he works.
Write down an expression, in terms of h, for the total cost, in dollars, for h hours of work.

\$ ..... **[1]**

(b) Given the formula

$$P = \frac{k^3 - m^2}{\sqrt{n}}$$

Work out the exact value of P when $k = -2$, $m = -3$ and $n = \frac{1}{16}$.

$P =$ ..... **[3]**

1 $3(4) - 2(-5)$ or $12 - -10$ or $12 + 10$ **[M1]**

22 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Substitute the given values into the formula: $v = 5 + (-2)(3)$ **[M1]**

$v = 5 - 6$ **[M1]**

-1 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 (a) $4(-2) - 3(5)$ or $-8 - 15$ **[M1]**

-23 **[A1]**

(b) $\frac{5^2 - 7}{2(-3)}$ or $\frac{25 - 7}{-6}$ **[M1]**

-3 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 (a) Substitutes values correctly: $5(3) - 2(-4)$ **[M1]**

$15 + 8 = 23$ **[A1]**

(b) Substitutes values: $\frac{(-4)^2 - (2)^3}{2(3)}$ **[M1]**

Evaluates numerator: $16 - 8 = 8$

$\frac{8}{6} = \frac{4}{3}$ or $1\frac{1}{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 (a) (a) $40 + 25(3.5)$ or $40 + 87.5$ **[M1]**

127.50 **[A1]**

(b) $-12 + 9.8 \times 3$ **[M1]**

$-12 + 29.4$ **[M1]**

17.4 **[A1]**

[Back to Question] *[Total Marks: 5]*

6 (a) $d = \sqrt{(-8 - 4)^2 + (2 - -3)^2}$ or $\sqrt{144 + 25}$ **[M1]**

13 **[A1]**

(b) $17 = \sqrt{(3 - k)^2 + (-7 - 8)^2}$ **[M1]**

$289 = (3 - k)^2 + 225$ **[M1]**

$(3-k)^2 = 64$ or $k^2 - 6k - 55 = 0$ **[M1]**

$k = -5$ and $k = 11$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** States $A =$ Curved Surface Area $+$ Base Area $= 2\pi rh + \pi r^2$ **[M1]**

Factorises πr out to obtain $\pi r(2h + r)$ **[A1]**

(b) **(i)** Substitutes $r = 4$ and $h = 6$: $A = \pi(4)(2(6) + 4)$ **[M1]**

$A = 4\pi(12 + 4) = 64\pi$ **[A1]**

(ii) Rearranges volume formula: $h = \dfrac{V}{\pi r^2}$ **[M1]**

Substitutes h into area formula: $A = 2\pi r\left(\dfrac{V}{\pi r^2}\right) + \pi r^2$ **[M1]**

Simplifies first term: $\dfrac{2\pi rV}{\pi r^2} = \dfrac{2V}{r}$, resulting in $A = \dfrac{2V}{r} + \pi r^2$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Substitute $x = -2$ and $y = 4$: $P = \dfrac{2(-2)^2 - 3(4)}{-2 - 4}$ **[M1]**

$P = \dfrac{2(4) - 12}{-6} = \dfrac{8 - 12}{-6}$ **[M1]**

$\dfrac{2}{3}$ (or equivalent) **[A1]**

(b) Multiply both sides by $(x - y)$: $P(x - y) = 2x^2 - 3y$ **[M1]**

Expand: $Px - Py = 2x^2 - 3y$ **[M1]**

Collect terms with y on one side: $3y - Py = 2x^2 - Px$ **[M1]**

Factorise and divide: $y(3 - P) = 2x^2 - Px \implies y = \dfrac{2x^2 - Px}{3 - P}$ (or equivalent) **[A1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** $V = \dfrac{1}{3}\pi(6.4)^2(12.6)$ **[M1]**

540.5 **[A1]**

(b) $800 = \dfrac{1}{3}\pi r^2(14)$ **[M1]**

$r^2 = \dfrac{2400}{14\pi}$ or $54.567\ldots$ **[M1]**

7.39 **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** Substitute $h = 3.5$: $C = 45 + 35(3.5)$ **[M1]**

167.5 **[A1]**

(b) Substitute $C = 325$: $325 = 45 + 35h$ **[M1]**

$280 = 35h$ **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 5]*

11 (a) Substitutes $h = 4.2$ and $a = 6.5$ into formula: $V = \frac{1}{6}\pi(4.2)(3(6.5)^2 + 4.2^2)$ **[M1]**

$V = 317.53... = 317.5$ **[A1]**

(b) Multiplies by 6 and divides by πh: $\frac{6V}{\pi h} = 3a^2 + h^2$ **[M1]**

Subtracts h^2: $\frac{6V}{\pi h} - h^2 = 3a^2$ **[M1]**

$a^2 = \frac{2V}{\pi h} - \frac{h^2}{3}$ or $a^2 = \frac{1}{3}\left(\frac{6V}{\pi h} - h^2\right)$ **[A1]**

(c) $a = \sqrt{\frac{2(150)}{\pi(3)} - \frac{3^2}{3}} = \sqrt{31.83... - 3} = \sqrt{28.83...} = 5.37$ **[B1]**

[Back to Question] *[Total Marks: 6]*

12 (a) (a) $5.2(4.5) + 0.5(-9.8)(4.5)^2$ **[M1]**

-75.825 **[A1]**

(b) $-9.8 = 4.9t + \frac{1}{2}(-9.8)t^2$ **[M1]**

$4.9t^2 - 4.9t - 9.8 = 0$ or $t^2 - t - 2 = 0$ **[M1]**

$t = 2$ (rejects -1 as time must be positive) **[A1]**

[Back to Question] *[Total Marks: 5]*

13 Substitution of all three values correctly e.g. $\frac{3(5) - (-4)^2}{2(0.5)}$ **[M1]**

-1 **[A1]**

[Back to Question] *[Total Marks: 2]*

14 (a) $85(7) + 0.24(450)$ **[M1]**

703 **[A1]**

(b) $1475.60 = 85(14) + 0.24m$ **[M1]**

$1475.60 - 1190 = 0.24m$ (or $285.60 = 0.24m$) **[M1]**

1190 **[A1]**

[Back to Question] *[Total Marks: 5]*

15 **(a)** **(i)** $\dfrac{6^2-(-2)^2}{6(-2)}$ **[M1]**

$-\dfrac{8}{3}$ **[A1]**

(ii) $\dfrac{0.5^2-4^2}{0.5(4)}$ **[M1]**

$\dfrac{-15.75}{2}$ **[M1]**

$-\dfrac{63}{8}$ **[A1]**

(b) $\dfrac{x^2-2^2}{2x}=\dfrac{3}{2}$ **[M1]**

$2(x^2-4)=6x$ (or $x^2-3x-4=0$) **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 8]*

16 Correct substitution: $5(-2)^2-3(4)$ or $5(4)-12$ **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 2]*

17 **(a)** $45+32h$ (or equivalent) **[B1]**

(b) 157 **[B1]**

(c) Correct substitution: $(-3)^3-4(-3)(2.5)$ or $-27-(-30)$ **[M1]**

3 **[A1]**

[Back to Question] *[Total Marks: 4]*

18 Numerator correctly evaluated: $-4.5-(-3.2)^2=-4.5-10.24=-14.74$ **[B1]**

Denominator correctly evaluated: $1.8-\sqrt{0.64}=1.8-0.8=1$ **[B1]**

-14.74 **[B1]**

[Back to Question] *[Total Marks: 3]*

19 **(a)** -6 or -20 seen **[M1]**

-26 **[A1]**

(b) $(-3)^2=9$ or $4(-3)=-12$ seen **[M1]**

-3 **[A1]**

[Back to Question] *[Total Marks: 4]*

20 $\pi \times (4.5)^2 \times 8.2 + \frac{2}{3} \times \pi \times (4.5)^3$ **[M1]**

$521.6... + 190.8...$ or $712.5...$ **[M1]**

713 **[A1]**

[Back to Question] *[Total Marks: 3]*

21 **(a)** $40 + 25h$ **[B1]**

(b) $(-2)^3 = -8$ and $(-3)^2 = 9$ **[M1]**

$\sqrt{\frac{1}{16}} = \frac{1}{4}$ **[B1]**

-68 **[A1]**

[Back to Question] *[Total Marks: 4]*